

A Bootstrap-Based Comparison of Portfolio Insurance Strategies

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Abstract

This study presents a systematic comparison of portfolio insurance strategies. In order to test for statistical significance of the differences in downside performance risk measures between pairs of portfolio insurance strategies, we use a bootstrap-based hypothesis test. Our comparison of different strategies considers the following distinguishing characteristics: static versus dynamic; initial wealth versus cumulated wealth protection; model-based versus model-free; and strong floor compliance versus probabilistic floor compliance. Our results show that the classical portfolio insurance strategies synthetic put and CPPI provide superior downside protection compared to a simple stop-loss trading rule, also resulting in significantly higher Omega ratios. Analyzing more recently developed strategies, neither the TIPP strategy (as an ‘improved’ CPPI strategy) nor the dynamic VaR-strategy provide significant improvements over the more traditional portfolio insurance strategies. The attractiveness of the dynamic VaR-strategy strongly depends on the quality of the estimates for the required input parameters, in particular, the equity risk premium. However, if an investor possesses superior forecasting skills, other active (market timing) strategies may exist which generate higher (risk-adjusted) returns compared to a protected passive stock market investment.

Keywords: Portfolio insurance strategies, Omega ratio, bootstrap simulation, Monte Carlo simulation

JEL classification codes: G11, G13

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I. Introduction

Portfolio insurance is of utmost importance for the asset management industry. In fact, driven by their return and risk preferences or by regulatory requirements, many investors are highly interested in preserving their invested capital. The major objective of any portfolio insurance strategy is to limit the downside risk of a risky asset (or a portfolio of risky assets), while simultaneously maintaining most of the upside return potential. Institutional investors often use portfolio insurance strategies in tailor-made solutions to protect their portfolios against large losses, whereas many private investors invest their capital in mutual funds that are endowed with a capital protection guarantee. The literature offers a number of different methods and strategies to secure risky assets against large losses, such as the stop-loss strategy, the synthetic put strategy, or the constant proportion portfolio insurance (CPPI) strategy. Therefore, once an investor has decided to use a protection strategy within his or her asset allocation framework, it may be of critical importance to implement the “best” strategy which fits the investor’s preferences as good as possible.

The introduction of synthetic put strategies by Rubinstein and Leland (1981) laid the foundation for numerous studies that contribute to explain why portfolio insurance is so popular in investment practice. Prior studies investigate the optimality of portfolio insurance strategies in a standard expected utility maximization setup (Leland, 1980; Brennan and Solanki, 1981; Benninga and Blume, 1985; Dybvig, 1988; Black and Perold, 1992).¹ A consensus in this strand of the literature is that portfolio insurance strategies are only utility maximizing under very specific assumptions with respect to investor preferences. In a more recent strand of the literature, Annaert et al. (2009) find that portfolio insurance strategies outperform a buy-and-hold strategy in terms of downside protection, but provide lower excess returns. A comparison based on stochastic dominance criteria reveals that no dominance relations can be identified between portfolio insurance strategies and buy-and-hold.² Given that their lower return

¹ Brennan and Schwartz (1989), Basak (1995), and Grossman and Zhou (1996) analyze the equilibrium implications of portfolio insurance in an expected utility framework.

² Annaert et al. (2009) implement block-bootstrap simulations based on the empirical return distribution to incorporate heavy tails and volatility clustering. Bertrand and Prigent (2005) and Zagst and Kraus (2011) also evaluate portfolio insurance performance using stochastic dominance. For example, Zagst and Kraus (2011) derive parameter conditions implying a second- and third-order stochastic dominance of the CPPI strategy against the protective put strategy. However, they assume that the underlying risky asset follows a geometric Brownian motion.

potential is sufficiently compensated by lower risk, Annaert et al. (2009) conclude that portfolio insurance strategies are valuable alternatives at least for some investors. Dichtl and Drobetz (2011) analyze portfolio insurance strategies within the framework of the cumulative prospect theory. Their simulation results indicate that loss aversion and probability weighting contribute to make most portfolio insurance strategies a preferred investment strategy for a prospect theory investor. Dierkes et al. (2010) similarly document that portfolio insurance is attractive for an investor with preferences described by the cumulative prospect theory.

Given the strategic decision to implement a portfolio insurance strategy, the question of interest is which particular strategy should be adopted. In fact, several prior studies evaluate and compare different pairs of portfolio insurance strategies in terms of their return and risk characteristics. For example, Benninga (1990) compares the stop-loss rule, the CPPI strategy, and the synthetic put strategy by using a Monte Carlo simulation approach. He reports that the simple stop-loss rule dominates both the CPPI strategy and the synthetic put strategy in terms of their terminal wealth and Sharpe ratio.³ Cesari and Cremonini's (2003) simulation results show that the relative performance of portfolio insurance strategies depends on the market environment. For example, using different performance measures (including alternative risk measures, such as the downside deviation and the Sortino ratio), the CPPI strategy dominates all other strategies in bear and sideways markets.⁴

Herold et al. (2005) examine a value-at-risk-based (VaR) dynamic investment strategy, where the protection floor is maintained with a pre-specified probability (probabilistic floor compliance). They compare this strategy with the CPPI strategy and document a higher return of the dynamic VaR-strategy in the context of fixed-income investments. Jiang et al. (2009) also compare a variant of a dynamic VaR-based portfolio insurance strategy with the CPPI strategy and buy-and-hold. Their results indicate that the dynamic VaR-based strategy outperforms the CPPI strategy in terms of downside protection and return potential. A caveat is that Jiang et al. (2009) approximate the expected return and volatility using their realized values during

³ Do (2002) compares the CPPI strategy and the synthetic put strategy using Australian market data. While neither strategy can be justified based on a loss minimization or a gain participation point of view, he nevertheless reports that the CPPI strategy dominates in terms of floor protection and the costs of insurance.

⁴ Bookstaber and Langsam (2000) also provide a detailed analysis of the path-dependency of portfolio insurance strategies.

the insurance period, thus their analysis neglects the estimation risk inherent in the dynamic VaR-strategy. Bertrand and Prigent (2011) argue that the Omega measure is the most adequate performance measure because it is able to capture the entire empirical return distribution of portfolio insurance strategies. Using a block-bootstrap simulation approach, they compare the CPPI strategy with the synthetic put strategy and report a dominance of the CPPI strategy in terms of the Omega ratio.

Our study complements and expands these earlier studies which evaluate and compare different portfolio insurance strategies. We not only analyze different pairs of portfolio insurance strategies in terms of their protection quality and return potential, but rather, provide a comprehensive statistical analysis of five protection strategies: the traditional stop-loss strategy, the synthetic put strategy, the CPPI strategy, the time invariant portfolio protection (TIPP) strategy, and the VaR-based protection strategy. Using a bootstrap simulation approach based on historical data as well as scenario-based Monte Carlo simulation, we investigate the choice between the following characteristics of different portfolio insurance strategies: (1) static versus dynamic protection; (2) initial wealth versus accumulated wealth protection; (3) model-based versus model-free protection; and (4) strong floor compliance versus probabilistic floor compliance. In each case, we compare two portfolio insurance strategies by using Keating and Shadwick's (2002) Omega ratio. By implementing a bootstrap-based hypothesis test which does not require specific assumptions about the distributional properties of the Omega ratios as well as all other applied performance and risk measures themselves, we are able to provide statistical inference. For those strategies which require parameter estimates (i.e., the synthetic put strategy and the dynamic VaR-strategy), we incorporate estimation risk.

Our results indicate that the 'classical' portfolio insurance strategies CPPI and synthetic put dominate all other strategies under realistic investment conditions. Both strategies are superior to the simple stop-loss trading rule in terms of their protection quality and overall risk-adjusted performance measured by the Omega ratio. Furthermore, recently developed portfolio insurance strategies also seem to be at a disadvantage against the more standard variants. Specifically, while the TIPP strategy (as an 'improved' CPPI strategy) exhibits a very limited return potential, the attractiveness of the dynamic VaR-strategy is strongly dependent on the parameters used to implement the strategy, in particular, the forecasting quality for the equity risk premium.

The remainder of this paper is structured as follows: Section II provides an overview and a brief discussion of the implemented portfolio insurance strategies. While section III describes our historical simulation methodology and our test design for statistical significance, section IV presents the historical simulation results. As a robustness test, section V documents the methodology and results of Monte Carlo simulations. Section VI concludes and emphasizes the implications for the investment practice.

II. An overview of portfolio insurance strategies

A variety of portfolio insurance strategies have been suggested in the prior literature, such as the stop-loss trading rule, the use of derivatives (e.g., a protective put strategy), and dynamic trading strategies (e.g., a synthetic put strategy or the CPPI strategy). In contrast to the stop-loss trading rule and dynamic trading strategies, a protective put strategy requires a liquid put option with the desired strike price and the desired time to maturity (Figlewski et al., 1993). Therefore, we omit option-based portfolio insurance strategies and focus on the simple stop-loss trading rule and dynamic trading strategies.⁵

A. Stop-loss portfolio insurance strategy

The simplest way to protect a risky portfolio against losses is the stop-loss portfolio insurance strategy (Bird et al., 1988). An investor invests his total wealth (W_0) in the risky asset. This position is maintained as long as the market value of the risky position (W_t) exceeds the net present value (NPV) of the floor (F_T), which is the minimum acceptable portfolio value at the end of the investment horizon T :

$$(1) \quad W_t > NPV(F_T).$$

If the portfolio value reaches or drops below the discounted floor, i.e., if $W_t \leq NPV(F_T)$, all risky portfolio holdings are sold and invested in the risk-free asset. This position is held until the end of the investment horizon. As long as the interim portfolio value does not drop below $NPV(F_T)$, an investor's final wealth will never be lower than F_T . The stop-loss portfolio insurance strategy is easy to implement, does not depend on any assumptions, and also does not

⁵ The following description of portfolio insurance strategies is based on Dichtl and Drobetz (2011).

require estimating any model parameters. However, the investor no longer participates from an upward market movement once the portfolio has been shifted in the risk-free asset.

B. Synthetic put portfolio insurance strategy

A second portfolio insurance strategy is Rubinstein and Leland's (1981) synthetic put strategy, which uses the Black-Scholes (1973) option pricing formula to create a continuously adjusted synthetic European put option on the risky asset. Combining the purchase of the risky asset with the purchase of a put on this asset (stock) is equivalent to purchasing a continuously-adjusted portfolio which is a combination of the risky asset and the risk-free asset (cash). The value of a portfolio that consists of a stock S plus a put P can be calculated as:

$$(2) \quad S + P = S - S \times N(-d_1) + K \times e^{-rT} N(-d_2) = \\ S \times [1 - N(-d_1)] + K \times e^{-rT} N(-d_2) = S \times N(d_1) + K \times e^{-rT} N(-d_2),$$

where K is the strike price, r the risk-free rate, and T the time to maturity. $N(\cdot)$ is the standard normal cumulative distribution function with d_1 and d_2 defined as:

$$(3a) \quad d_1 = \frac{\ln(S/K) + (r + 0.5\sigma^2)T}{\sigma\sqrt{T}} \quad \text{and} \quad (3b) \quad d_2 = d_1 - \sigma\sqrt{T},$$

where σ denotes the standard deviation of risky asset returns. In order to calculate the investment in the risky asset of the replicating portfolio, the delta of the portfolio in equation (2) is:

$$(4) \quad \frac{\partial(S+P)}{\partial S} = N(d_1).$$

Multiplying the delta with the price of the risky asset S and dividing by the value of the portfolio in equation (2), the percentage allocations (w) in the risky asset and the risk-free asset are given as:

$$(5a) \quad w_{risky} = \frac{S \times N(d_1)}{S \times N(d_1) + K \times e^{-rT} \times N(-d_2)} \quad \text{and} \quad (5b) \quad w_{risk-free} = 1 - w_{risky}.$$

This strategy requires increasing (decreasing) the proportion of the risky asset in the portfolio if the price of the risky asset increases (decreases). In order to maintain a desired protection level, the strike price K must be set such that the following relationship holds:

$$(6) \quad K = \frac{F_T}{w_0} \times (S + P(K)),$$

where the ratio F_T/W_0 is the percentage floor. The solution of equation (6) must be determined iteratively, because the value of the put option $P(K)$ depends on the strike price itself. Given that the put itself is costly, the exercise price K of the put will be higher than the floor.

The portfolio must be readjusted on a continuous basis in order to maintain the desired protection level. However, transaction costs will incur with each portfolio adjustment, and thus a higher adjustment frequency leads to higher transaction costs. In order to incorporate this transaction costs effect, Leland (1985) as well as Boyle and Vorst (1992) suggest using the synthetic put portfolio insurance strategy with a modified volatility estimator:

$$(7a) \quad \sigma_{Leland} = \sigma \sqrt{1 + \sqrt{\frac{2}{\pi} \times \frac{k}{\sigma \sqrt{\Delta t}}}} \quad (7b) \quad \sigma_{Boyle/Vorst} = \sigma \sqrt{1 + 2 \times \frac{k}{\sigma \sqrt{\Delta t}}}$$

where k captures the round-trip transaction costs, and Δt denotes the length of the readjustment period. With these volatility adjustments, both the readjustment frequency and the corresponding transaction costs are taken into account.

C. Constant proportion portfolio insurance strategy (CPPI)

In contrast to a synthetic put strategy, the CPPI strategy is a dynamic protection strategy that is not based on option pricing theory (Black and Jones, 1987). Starting point is an investor's risk capital at time t (the 'cushion'). The current cushion C_t represents the difference between current wealth at time t , labeled W_t , and the discounted floor $NPV(F_T)$:

$$(8) \quad C_t = W_t - NPV(F_T).$$

The exposure into the risky asset at time t , denoted as e_t , is calculated by multiplying the cushion C_t with the multiplier m :

$$(9) \quad e_t = m \times C_t.$$

The multiplier m could be set to any value, but it has a strong economic implication. Specifically, the inverse of the multiplier ($1/m$) represents the maximum sudden loss in the risky asset that may occur such that the cushion is not fully depleted and the portfolio value does not fall below the discounted floor. Although maintenance of the floor will be controlled on an intraday basis, the risk still exists over night when the portfolio manager cannot react im-

mediately to extreme market losses (overnight risk or gap risk).⁶ In spite of this small (unlikely) residual risk, which can partly be controlled by choosing a small multiplier m , the CPPI strategy can be classified as an absolute protection strategy with a strictly lower limit on the portfolio value. In commercial applications, the CPPI strategy is usually implemented such that the exposure of the risky asset varies between 0% and 100%. This implies that short sales and leverage are excluded (Benninga, 1990; Do, 2002; Annaert et al., 2009):

$$(10) \quad e_t = \max[\min(m \times C_t, W_t), 0].$$

The CPPI strategy can be implemented in our simulation analysis as dictated in equation (10) by shifting between stocks and bills.

D. Time invariant portfolio protection strategy (TIPP)

Estep and Kritzman (1988) argue that investors will not only be interested in the protection of their initial wealth, but also in locking in all interim capital gains. They suggest a modification of the standard CPPI strategy, which they call the ‘time invariant portfolio protection’ (TIPP) strategy. While the CPPI strategy operates with a fixed floor (i.e., the initial wealth multiplied by the percentage floor), the floor of the TIPP strategy is ratcheted up if the value of the portfolio increases. After choosing the floor and the multiplier, this strategy requires the following steps (Estep and Kritzman, 1988):

1. Calculation of the actual portfolio value (stock plus cash).
2. Multiplication of this portfolio value by the floor percentage.
3. If the result in step 2 is greater than the previous floor, it becomes the new floor; otherwise the old floor is kept.
4. Application of the CPPI strategy as dictated by equations (8)-(10).

As in the CPPI strategy, the TIPP strategy transfers all risky asset holdings in an irreversible manner to the risk-free asset once the floor has been reached. Therefore, the TIPP strategy cannot participate from subsequent upward market movements. Given the continuous ‘ratch-

⁶ For example, with a multiplier of $m = 5$, the risky asset can lose 20% ($1/5 = 0.20$) without violating the floor. When a sudden loss of over 20% occurs, the value of the portfolio falls below the promised minimum value (gambler’s ruin). In commercial applications it is necessary to continuously control the optimal exposure in the risky asset, thus portfolio shifts would have to be executed immediately. In most instances, however, an appropriate trading filter is used.

eting up' of the floor to the highest portfolio value, the likelihood that the portfolio value reaches or violates the prevailing floor increases, potentially implying adverse return effects because the TIPP strategy may more often end up fully invested in the risk-free asset (Choi and Seff, 1989; Dichtl and Drobetz, 2011).

E. Dynamic value-at-risk portfolio insurance strategy

The goal of the dynamic VaR-based strategy is to control the exposure of a risky asset such that a specified value at risk is not violated (Jiang et al., 2009). The value-at-risk (*VaR*) of any portfolio P defines the maximum loss that will not be exceeded based on a defined confidence level within a given time period:

$$(11) \quad VaR = T \times \mu_P + \tau_{1-\alpha} \times \sqrt{T} \times \sigma_P,$$

where T is the time horizon and $\tau_{1-\alpha}$ the $(1 - \alpha)$ -quantile of the standard normal distribution. μ_P and σ_P denote the expected portfolio return and volatility, respectively. Consider a risk-free position (with return r_f) and a stock (market) position (with expected return μ_S and volatility σ_S), where x represents the percentage stock market allocation. Given that the risk-free asset has zero volatility and is uncorrelated with stock returns, the value-at-risk is derived as follows:⁷

$$(12) \quad VaR = T \times [x \times (\mu_S - r_f) + r_f] + \tau_{1-\alpha} \times \sqrt{T} \times x \times \sigma_S.$$

Within this VaR-based asset allocation, the question is how to allocate the portfolio wealth between stocks and cash in order to ensure that a given VaR-limit will not be violated. More specifically, the value-at-risk in equation (12) is set to a pre-specified value, and the expression is solved for the stock market allocation x :

$$(13) \quad x = \frac{VaR - T \times r_f}{T \times (\mu_S - r_f) + \tau_{1-\alpha} \times \sqrt{T} \times \sigma_S}.$$

⁷ In our implementation of the dynamic VaR-strategy, we perform all calculations based on continuously compounded returns. In order to calculate the continuously compounded portfolio return as a weighted average of the stock market and the cash market position, we transform the continuously compounded stock and cash market returns into simple returns. The simple portfolio return is then transformed back into a continuously compounded return for all following calculations. These elementary transformations are not incorporated in equation (12) for the purpose of simplification.

Various input parameters must be specified in equation (13): the confidence level α , the risk-free rate r_f , the expected stock market volatility σ_S , and the equity risk premium $(\mu_S - r_f)$. Both the future stock market volatility and the expected risk premium are ex ante unobservable and need to be estimated. Instead of applying equation (13) only once at the beginning of the insurance period, an investor needs to continuously readjust the stock-cash allocation. In this sense, the VaR-strategy constitutes a dynamic trading strategy comparable to the CPPI or the synthetic put strategy.⁸ Commercial applications avoid leverage and short sales, and thus the percentage stock market allocation x in equation (13) must vary between 0% and 100% in our simulation analysis.

F. Distinguishing characteristics of portfolio insurance strategies

The main properties of the different portfolio insurance strategies discussed in this section are summarized in Table 1. First of all, portfolio insurance strategies can be classified into static and dynamic strategies. While static strategies maintain their initial asset allocation or change it only once during the entire insurance period, dynamic protection strategies adjust their allocation continuously on the basis of market movements. Therefore, the stop-loss trading rule is classified as a static strategy, while all other trading rules are dynamic strategies. A second distinguishing feature is the protection level. While the TIPP strategy (as a modified CPPI strategy) locks in the cumulated wealth, all other strategies set their protection target at the initial portfolio wealth. A third criterion for differentiation is the type of insurance provided. In contrast to the strategies stop-loss, synthetic put, CPPI, and TIPP, a dynamic VaR-strategy maintains the floor only with a pre-specified probability. We refer to this feature as ‘probabilistic floor compliance’ as opposed to ‘strong floor compliance’ of all other portfolio insurance strategies in Table 1. The fourth criterion for differentiation relates to the model parameters. While some strategies require one or two model parameters (stop-loss, CPPI, and TIPP), other strategies depend on more model parameters (synthetic put and dynamic VaR-strategy). Furthermore, both the synthetic put and the dynamic VaR-strategy include forward looking

⁸ It is important to consider cumulated portfolio performance when implementing the dynamic VaR-strategy. For example, if the target-VaR over a one-year investment horizon is 0% and the portfolio wealth has increased from 100 to 110 after one month, the VaR-budget is -9.09%, thus a loss of up to -9.09% does not violate the original VaR-budget. In contrast, if the portfolio wealth has decreased from 100 to 90, the new VaR-budget becomes +11.11%. The portfolio value must increase by at least 11.11% in order to maintain the original VaR-budget of 0%.

model parameters (Bird et al., 1990; Zhu and Kavee, 1988; Jiang et al., 2009), and therefore the precision of these parameter estimates will impact the quality of loss protection (Rendleman and O'Brien, 1990). Finally, the various strategies can be differentiated in terms of their model assumptions. While stop-loss, CPPI, and TIPP do not depend on any model assumptions, the synthetic put strategy and the dynamic VaR-strategy are based on normally distributed returns.

[Insert Table 1 here]

Taken together, our systematic comparison of the different portfolio insurance strategies is structured along the following distinguishing characteristics:

- *Static versus dynamic portfolio insurance:* In a first step of our simulation analysis, we compare the simple stop-loss trading rule with the CPPI strategy (comparison 1). Apart from the difference between static and dynamic protection, the two strategies are equal along all other dimensions. Both strategies focus on the protection of initial wealth, provide strong floor compliance, are free of any model assumptions, and do not require the estimation of model parameters.
- *Initial wealth versus cumulated wealth protection:* Second, we compare the CPPI strategy with the related TIPP strategy (comparison 2). The latter TIPP strategy is a modified CPPI strategy, where the only difference relates to the level of wealth protection. Both strategies are dynamic, guarantee a strong floor compliance, are independent from any model assumptions, and do not require the forecast of any model parameters.
- *Model-based versus model-free portfolio insurance:* Third, to assess model-based against model-free protection strategies, we compare the synthetic put with the CPPI strategy in a third step (comparison 3). While both strategies are dynamic, their main difference relates to the model assumptions. In contrast to the CPPI strategy, the synthetic put strategy is derived from the Black and Scholes (1973) option pricing model and depends on its underlying assumptions. Moreover, the synthetic put trading rule requires estimating the stock market volatility to be used in the option pricing formula.
- *Strong floor compliance versus probabilistic floor compliance:* In a final step, we examine the dynamic VaR-strategy in comparison to the synthetic put strategy (comparison

4a). Both strategies are dynamic, assume normally distributed stock returns, and require an estimation of unobservable strategy parameters. Furthermore, we compare the dynamic VaR-strategy with the CPPI strategy (comparison 4b). While both strategies are dynamic, the CPPI strategy is independent of any model assumptions and parameter estimates. Taken together, comparing these two pairs of strategies highlights the difference between strong floor compliance (synthetic put and CPPI strategy) and probabilistic floor compliance (dynamic VaR-strategy).

III. Historical simulation methodology

A. Data and design of the empirical analysis

Our data consists of daily returns for the German stock market index DAX and money market rates from the German Bundesbank (average values of the 1-month Frankfurt interbank rate; middle rate) over the sample period from January 1980 to December 2011.⁹ As documented in Poterba and Summers (1988), among others, stock market returns exhibit short-term autocorrelation and long-term mean reversion. Moreover, stock market returns deviate from the normal distribution with constant volatility (homoskedasticity); they tend to be heteroskedastic and left-skewed, and exhibit fat tails (Annaert et al., 2009; Bertrand and Prigent, 2011).

[Insert Table 2 here]

Panel A in Table 2 presents descriptive statistics of the continuously compounded daily stock market returns and money market rates. With an annual mean return of 7.38% the German stock market exhibits a risk premium of 2.68% during our sample period against the annual money market rate of 4.70%. With an annual value of 21.53% the German stock market features a standard deviation (volatility) that can be considered as normal (see section V.). Panels B-D of Table 2 illustrate the statistical properties of the continuously compounded daily DAX returns. Panel B shows the autocorrelations. A Ljung-Box test up to lag five indicates statistically significant autocorrelations, albeit all values are close to zero. In order to analyze heteroskedasticity effects, we implement Engle's (1982) Lagrange multiplier test. The results in Panel C indicate statistically significant heteroskedasticity effects in daily DAX returns at

⁹ Average values are calculated as the mean values of all interbanking lending transactions in the Frankfurt marketplace.

lag one and up to lag three. Furthermore, the daily DAX returns are left-skewed (skewness of -0.32) and exhibit fat tails, as shown by the value of 10.03 for the kurtosis. The corresponding Jarque-Bera test statistic in Panel D reveals that the null hypothesis of a normal distribution must be rejected at the 1% significance level. Overall, our daily DAX returns match the typical stock market properties quite well.

In order to fully incorporate all the time series characteristics, we follow Annaert et al. (2009) and Bertrand and Prigent (2011) and implement a block-bootstrap by drawing 250 subsequently following pairs of daily stock and money market returns.¹⁰ Selecting $N = 5,000$ coherent data blocks with replacement and applying the different portfolio insurance strategies in each draw provides us with 5,000 annual returns for each strategy.¹¹ The $N = 5,000$ yearly returns are the basis for our performance measures (see section III.B) as well as our hypothesis tests (see section III.C).

In our baseline simulations, the equity risk premium and the volatility for each one-year simulation path of the dynamic VaR-strategy are derived from the prior 250 daily returns. Therefore, the sample period for all of our strategy simulations is from January 1981 to December 2011. In order to assess the impact of alternative return estimates, we also estimate the stock market return (used to calculate the equity risk premium) as an average over our full sample period (7.38% per year) as well as the long-run stock market return (9% per year; based on the stock market data collected by Dimson et al., 2006).¹² Furthermore, we put the equity risk premium to zero, which is often done for VaR-calculations in (short-term) market risk applications. To model risk differently, we estimate the stock market volatility using the exponentially-weighted moving average model and the GARCH(1,1) model. In addition, we simply calculate the volatility as the average of the squared returns (setting the mean value to zero).¹³ The return volatility for the synthetic put strategy (estimated from the prior 250 daily returns)

¹⁰ Benartzi and Thaler (1995) provide evidence that institutional and private investors tend to use a one-year investment horizon. Therefore, we focus on this time period in our simulation analysis.

¹¹ As a robustness check, we also implement a circular block-bootstrap to avoid an underweighting at the beginning and the end of the original time series. Our results remain qualitatively the same.

¹² See Fama and French (2002) and Welch and Goyal (2008) for conceptual problems in estimating the equity risk premium.

¹³ A description of these different volatility estimation methods is shown in Hull (2009). Reasons for using the simple GARCH(1,1) model instead of more complicated GARCH models are discussed in Jacobsen and Dannenburg (2003).

is modified according to Boyle and Vorst (1992), as shown in equation (7b).¹⁴ We implement all strategies without short sales and leverage and take round-trip transaction costs of 10 basis points into account (Herold et al., 2007). In order to avoid portfolio shifts that are triggered by trendless market movements, we implement the dynamic VaR-strategy, the CPPI strategy, the TIPP strategy, and the synthetic put strategy with a trading filter. Portfolio shifts are only executed when the stock market moves by more than 2% (Do and Faff, 2004). Moreover, in the basic model specification, we implement both the CPPI strategy and the TIPP strategy with a multiplier of $m = 5$, which is commonly used in commercial applications (Herold et al., 2007). The confidence level in the dynamic VaR-strategy is $\alpha = 90\%$. We implement all strategies based on a protection level of 100% (i.e., with a full capital guarantee) and compare their performance with an unprotected buy-and-hold stock market investment.

B. Performance measures

In order to compare the different portfolio insurance strategies as described in section II.A-E, we investigate their protection quality and return potential on the basis of the $N = 5,000$ yearly returns. By using downside risk measures, we are able to appropriately quantify the protection quality of any given strategy. In particular, we compute lower partial moments (LPMs), which take only negative deviations from the pre-specified target return, denoted as τ , into account. As our analysis focuses on preserving the invested capital, the target return is set equal to zero ($\tau = 0$), and we penalize negative deviations from this target with an exponent of 0 (LPM0) and 1 (LPM1). LPM0 and LPM1 represent the shortfall probability and the expected shortfall, respectively (Harlow and Rao, 1989; Eling and Schuhmacher, 2007).¹⁵ We also report the empirical VaR (using a confidence level of $\alpha = 95\%$), the skewness, and the minimum annual return as alternative downside risk measures.

Following Bertrand and Prigent (2011), we use the Omega measure (Keating and Shadwick, 2002) for our systematic comparison of the different portfolio insurance strategies along the distinguishing characteristics as discussed in section II.F (comparisons 1-4). The Omega ratio represents a special case of the more general Kappa measure (Kaplan and Knowles, 2004). It

¹⁴ We also run the simulations using the original volatility or the Leland (1985) modification in equation (7a). The simulation results are not sensitive to these changes.

¹⁵ In the specific case of a zero target return, the LPM0 measure indicates the loss probability and the LPM1 measure the expected loss.

is characterized by the ratio of gains to losses relative to a predefined target return required by the investor:

$$(14) \quad \Omega(\tau) = \frac{\int_{\tau}^{\infty} (1-F(R))dR}{\int_{-\infty}^{\tau} F(R)dR},$$

where $F(\cdot)$ denotes the cumulative distribution function of returns R , and τ the target return (or threshold). The Omega ratio takes account of the entire return distribution, does not require any parametric assumptions, and can easily be computed by dividing the higher partial moment of degree 1 (HPM1) by the lower partial moment of the same degree (LPM1).¹⁶ In order to get a reasonable ratio of gains to losses, the target return should be lower than the expected portfolio return. In the context of portfolio insurance strategies, the threshold must also be higher than the protection level. Following Bertrand and Prigent (2011), we set the threshold for the Omega ratio to 1%, 2%, 3%, and 4% per year, respectively.

C. Testing for statistical significance

The starting point for our statistical significance tests are the $N = 5,000$ annual portfolio returns for each simulated protection strategy (based on the block-bootstrap approach from the first step). For all comparisons 1-4 formulated in section II.F, we evaluate the difference between the performance/risk measures (Π) of a pair of strategies A and B:¹⁷

$$(15) \quad \Delta_{\Pi} = \Pi_A - \Pi_B.$$

The objective is to test whether there is a significant difference between the performance/risk measures of the two compared strategies. Formally, we test:

$$(16) \quad H_0: \Delta_{\Pi} = 0 \quad \text{against} \quad H_1: \Delta_{\Pi} \neq 0.$$

¹⁶ The Omega ratio was used in prior portfolio insurance studies (for example, Bertrand and Prigent, 2011), thus enabling a high degree of comparability of our simulation results. There exist other performance measures that are also appropriate for the evaluation of portfolio insurance strategies, such as the upside potential ratio (Sortino et al., 1999) or the Farinelli-Tibiletti ratio (Farinelli et al., 2008). Apart from the threshold return, the Omega ratio does not require the specification of any other parameters. In contrast, the Farinelli-Tibiletti ratio requires the specification of two additional parameters which model an investor's attitude toward the consequences of extreme (catastrophic) events.

¹⁷ A similar type of a bootstrap-based statistical inference framework is implemented in Zeisberger et al. (2007).

Given the $N = 5,000$ annual returns for each strategy, an appropriate point estimator for the difference in equation (15) can be calculated:

$$(17) \quad \hat{\Delta}_{\Pi} = \hat{\Pi}_A - \hat{\Pi}_B.$$

However, a statistical test requires the distribution of the $\hat{\Delta}_{\Pi}$ value. Therefore, we implement a further bootstrap using the $N = 5,000$ annual returns from the block-bootstrap in the first step. Within this second step bootstrap, we create 5,000 bootstrap resamples ($N_B = 5,000$), each consisting of 500 annual returns.¹⁸ For each of the $N_B = 5,000$ bootstrap resamples, we compute $\hat{\Delta}_{\Pi}^*$ in the same way as for the point estimator $\hat{\Delta}_{\Pi}$ in equation (17). Furthermore, we define:

$$(18) \quad \hat{\Delta}_{\Pi[1]}^* \leq \hat{\Delta}_{\Pi[2]}^* \leq \dots \leq \hat{\Delta}_{\Pi[N_B]}^*$$

as the ordered series of the differences in the performance/risk measures. Based on this series a confidence interval can be constructed as described in Efron and Tibshirani (1998):

$$(19) \quad CI = [\xi_{low}^*; \xi_{high}^*],$$

where:

$$(20a) \quad \xi_{low}^* = \hat{\Delta}_{\Pi[\alpha/2 \times N_B]}^* \quad \text{and} \quad (20b) \quad \xi_{high}^* = \hat{\Delta}_{\Pi[(1-\alpha/2) \times N_B]}^*.$$

The null hypothesis H_0 is rejected at significance level α if $0 \notin CI$. We implement Efron's (1979) standard bootstrap technique because our bootstrap sample with $N = 5,000$ elements does not exhibit statistically significant dependencies. Independency is due to the fact that we obtain the 5,000 yearly returns of the different portfolio insurance strategies by randomly drawing blocks of 250 daily stock market and cash market returns with replacement.¹⁹

Figure 1 graphically summarizes our methodology, which consists of a block-bootstrap for the simulation of the annual returns of the respective portfolio insurance strategy in the first

¹⁸ Our bootstrap results are stable with this number of simulation runs.

¹⁹ If we implemented a rolling window approach instead of a bootstrap simulation, the resulting return series would suffer from an extremely high degree of serial dependency. In this situation, Efron's (1979) bootstrap methodology (which is based on the i.i.d. assumption) is not adequate. The stationary bootstrap technique proposed by Politis and Romano (1994) – a bootstrap technique which explicitly deals with weakly-dependent time series – is also no longer adequate. However, a comparison of our bootstrap simulation results with rolling window simulation results (not shown) confirms that both methodologies provide qualitatively the same results.

step and the traditional bootstrap of Efron (1979) for conducting the hypothesis tests in the second step. Specifically, let $R_i = (R_{Ai}, R_{Bi})$ be a pair of the annual returns of strategy A and B in the second step of our analysis. The starting point are the N pairs of annual returns, denoted as $(R_1, \dots, R_N)$, where we calculate the point estimator $\hat{\Delta}_{\Pi} = \Delta_{\Pi}(R_1, \dots, R_N)$. Similarly, we calculate the $\hat{\Delta}_{\Pi}^*$ values as $\hat{\Delta}_{\Pi}^* = \Delta_{\Pi}(R_1^*, \dots, R_{500}^*)$, where the R_i^* values refer to the results from the second-stage bootstrap (statistical inference bootstrap).

[Insert Figure 1 here]

IV. Historical simulation results

A. Main simulation results

The results of our bootstrap simulations are summarized in Panel A of Table 3. As expected, all portfolio insurance strategies generate positive skewness, which is in contrast to the negative skewness of the aggregate stock market return series. On average, the stop-loss strategy provides the highest mean annual return, but also the highest risk. In terms of the average annual return, both the synthetic put and the dynamic VaR-strategy also dominate the CPPI and the TIPP strategy. In contrast, the CPPI strategy and the TIPP strategy dominate all other strategies in terms of their risk, as indicated by the LPM0 and the LPM1 measures, the empirical VaR (at the 95% confidence level), and the minimum annual return (as our measure for protection quality). Finally, the Omega ratios (as our risk-adjusted performance measure) are reported for threshold returns ranging from 1% to 4%.

[Insert Table 3 here]

In order to implement our systematic strategy comparison, we assess the portfolio insurance strategies based on the differences in their performance and risk measures. Most important, a bootstrap-based hypothesis test as described in section III.C is applied to test the differences in the performance and risk measures for each pair of strategies (see section II.F). Panel B of Table 3 shows the differences between the performance and risk measures as well as the corresponding level of significance. Comparing the simple stop-loss trading rule with the CPPI strategy (comparison 1), we observe a clear dominance of the CPPI strategy in terms of the downside risk measures and the Omega ratio. However, the difference in the Omega ratio in

Panel B of Table 3 shrinks with increasing threshold returns, which is due to the lower return potential of the CPPI strategy in comparison to the stop-loss strategy. With a threshold return of $\tau = 4\%$, the difference in the Omega ratios is no longer statistically significant.

Comparing the stop-loss strategy with the other strategies in terms of their shortfall probability (LPM0), we generally observe a clear inferiority of the stop-loss trading rule. An explanation for this observation can be that the floor is breached almost immediately after the start of the strategy. Given the high volatility of stock returns, the resulting shortfall may be so large that the risk-free investment never compensates this loss. In order to analyze this observation in more detail, we run simulations of the stop-loss strategy with lower floor values. A lower floor may reduce the shortfall probability and enhance the risk-adjusted performance of the stop-loss strategy.²⁰ The results in Table 4 show that reducing the floor leads to lower shortfall probabilities and higher returns. At the same time, however, a reduction of the floor is accompanied by an increase of the expected shortfall (LPM1) measure. The comparison of the Omega ratios indicates that the reduction of the floor leads to a decreasing risk-adjusted performance of the stop-loss strategy. Our results are opposite to those in Benninga (1990), who finds a dominance of the stop-loss strategy in terms of risk-adjusted returns. He does not use downside risk measures and implements Monte Carlo simulations, potentially neglecting the properties of financial markets data (e.g., heteroskedasticity and fat tails).

[Insert Table 4 here]

In a second step, we compare the CPPI strategy and the TIPP strategy (comparison 2). Panel B of Table 3 reveals a statistically significant dominance of the TIPP strategy in terms of the Omega ratio with threshold returns up to 2%. Using a threshold return of 3%, the difference is no longer statistically significant. In contrast, when increasing the threshold return to 4%, the CPPI strategy provides a significantly higher Omega ratio compared to the TIPP strategy. This reversal is explained by the tradeoff between downside protection and return potential. Although the TIPP strategy exhibits a slightly superior downside protection (as indicated by the LPM0 and the LPM1 measure), it also provides a lower return potential. In order to assess whether these properties are specific for the chosen multiplier of $m = 5$ in our base-case sce-

²⁰ We are grateful to an anonymous referee for suggesting this additional analysis.

nario, we run additional bootstrap simulations and compare the CPPI strategy and the TIPP strategy with multipliers of $m = 3$, $m = 7$, and $m = 9$. The results are reported in Table 5; they generally confirm those in Table 3 above. While the TIPP strategy exhibits slightly superior risk measures, the CPPI strategy dominates in terms of annual mean returns. With high multipliers of $m = 7$ and $m = 9$, the mean return of the TIPP strategy does not increase notably, and the CPPI strategy leads to higher Omega ratios than the TIPP strategy when using threshold returns of 3% and 4%. The limited return potential of the TIPP strategy is a result of its property to protect the cumulated wealth. Given the continuous ratcheting up of the floor to the highest portfolio value, the higher floor is more often violated, implying a full cash investment until the end of the investment horizon and thus lower portfolio returns. Arguably, this effect will reinforce itself with increasing investment horizons. Overall, our results confirm Choie and Seff's (1989) general criticism of the TIPP strategy.

[Insert Table 5 here]

In a third step, we compare the synthetic put strategy and the CPPI strategy (comparison 3). As shown in Panel A of Table 3, the synthetic put strategy dominates the CPPI strategy in terms of mean annual returns. However, the synthetic put strategy provides worse downside protection, as indicated by the values of the LPM0 and LPM1 measures, the empirical VaR, and the minimum return. Based on the performance of the Omega ratio, we observe a dominance of the CPPI strategy against the synthetic put strategy for threshold returns up to 3% in Panel B of Table 3 (albeit only statistically significant for threshold returns of 1% and 2%). In contrast, using a threshold return of 4%, the synthetic put strategy is superior to the CPPI strategy in terms of Omega ratios at the 1% significance level. One explanation for this reversal is the choice of the multiplier. As shown in our robustness tests in Table 5, the return potential of the CPPI strategy increases with higher multipliers m . Moreover, the relatively poor protection quality of the synthetic put strategy may be explained by estimation risk for stock market volatility. Both issues are addressed in section IV.B in more detail.

In a fourth step, we evaluate the dynamic VaR-strategy against both the synthetic put and the CPPI strategy (comparison 4). The dynamic-VaR strategy shows a lower return potential than the synthetic put strategy, but it also generates slightly superior values for the downside risk measures in Panel A of Table 3. Moreover, based on the Omega ratio, we observe a domi-

nance of the synthetic put strategy for threshold returns from 2% to 4% (albeit only significant for the 4% threshold level). The results are opposite for comparisons of the VaR-strategy with the CPPI strategy. While the VaR-strategy provides higher mean annual returns, the CPPI strategy features superior downside protection properties. For threshold returns up to 3%, the Omega ratio of the CPPI strategy is significantly higher than that of the dynamic VaR-strategy.

A caveat is that the dynamic VaR-strategy is implemented using a fixed confidence level of $\alpha = 90\%$ (see equation 11), and it is not obvious whether this is the optimal choice. In order to evaluate the influence of the confidence level, we repeat our bootstrap simulations for varying values of the parameter α in a dynamic VaR-strategy, ranging from 0.85% to 0.99%. The results are summarized in Table 6. As expected, a reduction in the level of significance from 15% to 1% leads to superior downside protection, but also to lower returns. While the dynamic VaR-strategy with $\alpha = 0.85$ shows the highest mean annual return, the corresponding downside protection tends to be worse compared to model specifications with higher significance levels. The dynamic VaR-strategy based on $\alpha = 0.95$ appears particularly interesting. Comparing this strategy with the $\alpha = 0.99$ specification, we observe a higher mean annual return. At the same time, this strategy dominates both the $\alpha = 0.85$ and the $\alpha = 0.90$ specification in terms of the LPM0 and the LPM1 measure as well as the empirical VaR. In addition, based on Omega ratios, we find a dominance of the $\alpha = 0.95$ specification for all threshold return levels ranging from 1% to 4%.

[Insert Table 6 here]

The return-risk characteristics of a dynamic VaR-strategy not only depend on the confidence level α , but also on the forecast quality of the estimators for the risk premium and the volatility. In order to verify the impact of alternative estimators, we estimate the volatility using the exponentially-weighted moving average model and the GARCH(1,1) model (instead of using the volatility from the prior 250 daily returns).²¹ In addition, we simply calculate the volatility as the average of the squared returns (setting the mean value to zero). The results in Panel A of Table 7 indicate that alternative risk estimators do not lead to better risk-adjusted perfor-

²¹ In the exponentially-weighted moving average model, we set the decay factor λ ($0 < \lambda < 1$) to 0.94 (J.P. Morgan/Reuters, RiskMetrics™ – Technical Document, 1996).

mance. In fact, the VaR-strategy based the historical volatility model (in the last column) still dominates all other models in terms of the Omega ratio for all four threshold returns. While both the exponentially-weighted volatility model and the GARCH(1,1) model lead to slightly higher returns compared to the historical volatility model, this effect is overcompensated by a higher risk.

[Insert Table 7 here]

In order to use alternative risk premium estimates, we estimate the stock market return as the average return over our sample period (7.38% per year; see Table 2) and as the (very) long-run stock market return (9% per year) estimated by Dimson et al. (2006). In another robustness test, we set the equity risk premium to zero. The results in Panel B of Table 7 reveal that these alternative return estimators lead to a noticeable improvement of the risk-adjusted performance of the VaR model. For example, when we implement the VaR model with a fixed long-run return estimate of 9% per year, the Omega ratios are higher for all levels of threshold returns compared to the base-case scenario in the last column (for example, 19.73 versus 11.07 for $\tau = 1\%$). A comparison of our other performance measures indicates that the improvement is not attributable to the return side, but the risk side. While the VaR model with a fixed long-run return estimate of 9% per year provides a lower mean return compared to the base-case specification (6.70 versus 6.95 percent), this strategy dominates in terms of all risk measures. Setting the equity risk premium to zero also leads to an improvement of the applied risk measures.

B. Robustness tests

In this section, we conduct a series of robustness tests. Given the results in Tables 5 and 6, we start by repeating our bootstrap simulations with alternative parameterizations of the CPPI strategy and the dynamic VaR-strategy. The CPPI strategy is implemented with a multiplier of $m = 9$.²² The confidence level for the dynamic VaR-strategy is set to $\alpha = 0.95\%$, and the

²² In order to enable a comparison between the CPPI and the TIPP strategy, we also set the multiplier for the TIPP strategy to $m = 9$. The deviations in the performance measures for both strategies with $m = 9$ in Tables 5 and 8 are due to path dependency. Similar deviations across tables are observable for the stop-loss strategy and the dynamic VaR-strategy. However, all deviations are very small, indicating that our simulations exhibit sufficiently accurate convergence properties.

risk premium for this strategy is estimated using the fixed long-run stock market return of 9% (Dimson et al., 2006). The results are summarized in Table 8. With a more aggressive multiplier (compared to $m = 5$ in the base-case), the CPPI strategy again strongly dominates the simple stop-loss trading rule in terms of Omega ratios (comparison 1). The difference is statistically significant for all threshold returns ranging from 1% to 3% (Panel B of Table 8). The comparison of the CPPI and the TIPP strategy using a multiplier of $m = 9$ also confirms our earlier results (comparison 2). Again, the CPPI strategy generates higher Omega ratios compared to the TIPP strategy using threshold returns of 3% and 4%; the difference is statistically significant at a level of 1% in the latter case.

[Insert Table 8 here]

Comparing the CPPI strategy with the synthetic put strategy (comparison 3), our earlier results change. With the higher multiplier of $m = 9$, the CPPI strategy is dominated by the synthetic put strategy in terms of Omega ratios even by using a threshold return of 3%. While the synthetic put strategy provides higher mean returns than the CPPI strategy, the latter still offers superior downside protection. A comparison of the dynamic VaR-strategy with the synthetic put strategy again confirms earlier findings (comparison 4). The dynamic VaR-strategy exhibits superior values for the LPM0 and the LPM1 measure as well as the empirical VaR, but it also shows a lower return potential compared to the synthetic put strategy. In terms of the Omega ratio, the dynamic VaR-strategy significantly dominates the synthetic put strategy for thresholds of 1% to 3%. Comparing the Omega ratios of the dynamic VaR-strategy and the CPPI strategy, the dynamic VaR-strategy is now superior for all threshold return levels (significant at the 1% level). Accordingly, our results in Table 8 indicate that the advantage of the CPPI strategy against the dynamic VaR-strategy (as shown in Table 3) vanishes when a higher multiplier for the CPPI strategy is chosen (implying higher risk) and when the dynamic VaR-strategy is implemented using a (fixed) long-run average stock market return.

In contrast to all other strategies, the synthetic put strategy and the dynamic VaR-strategy are both dependent on the parameter forecasts. For example, comparing the LPM0 measures with a given parameter α for all different specifications of the dynamic VaR-strategy in Tables 3, 6, and 8, respectively, substantial differences emerge. In all cases, the LPM0 measure of the dynamic VaR-strategy indicates that the protection level is more often violated than expected

given the confidence level α . Furthermore, the synthetic put strategy exhibits a worse capital protection compared to other forecast-free strategies, such as the CPPI strategy and the TIPP strategy. While deviations from the assumption of normally distributed stock market returns may be one plausible explanation, the limited prediction quality of our volatility and risk premium estimates may offer another explanation. In order to further examine this issue, we repeat our simulations for the dynamic VaR-strategy and the synthetic put strategy. In contrast to our base-case simulations, where the volatility and risk premium estimates are taken from the year prior to the implementation of the strategy, we avoid estimation risk and apply the realized volatility and the realized risk premium in any given sample year. The results are shown in Table 9. Comparing the LPM0 measures of the dynamic VaR-strategy with and without estimation risk in Table 6 and Table 9, respectively, we observe a reduction in the LPM0 value towards the expected value as indicated by the confidence level α . For example, with $\alpha = 0.99$, the value of the LPM0 measure decreases from 10.44% to 2.20% when estimation risk is fully eliminated.

[Insert Table 9 here]

Another important effect is observed when the results of the dynamic VaR-strategy are compared for increasing confidence levels α . On the one hand, the protection quality is improved when the confidence level is increased. This expected result is indicated by the values of the LPM0 and LPM1 measures, the empirical VaR, and the minimum return. On the other hand, we also observe an increase in the mean annual return with increasing confidence levels (from 4.96% for $\alpha = 0.85$ to 7.69% for $\alpha = 0.99$). A more conservative dynamic VaR-strategy not only reduces risk, but it also enhances the return potential under ‘perfect’ forecasts.²³ These results suggest that both the protection quality and the return potential of the dynamic VaR-strategy can be enhanced if a manager possesses superior forecast skills and if a high confidence level is chosen. Unfortunately, these suggestions are only of limited relevance for the investment practice. The main reason for the application of portfolio insurance strategies is to protect a portfolio against negative market developments in every market environment, independent of whether market declines can be forecasted or not. Moreover, if an

²³ This effect is not found for the synthetic put strategy. Perfect volatility forecasts improve the protection quality of the synthetic put strategy, but returns cannot be enhanced (see Tables 3 and 9).

investor possesses superior forecasting skills, other active (market timing) strategies may exist which generate higher returns compared to a protected passive stock market investment. Taken together, our analysis weakens the results in Jiang et al. (2009). Based on our findings, we conjecture that the reason for the strong superiority of the dynamic VaR-strategy in their study compared to the CPPI strategy are the perfect quarterly return and volatility forecasts, which they use to implement their strategy. Our simulation analysis indicates that – once estimation risk is taken into account – the attractiveness of the dynamic VaR-strategy strongly depends on the parameter specification.

V. Monte Carlo simulations

A. Design of the analysis

In this section, we check the robustness of our bootstrap results by additionally running Monte Carlo simulations. This approach allows us to analyze different economic scenarios with different stochastic parameters. The two main parameters that are required for the simulation of stock market returns are the expected return and its corresponding volatility. We define four states of nature to analyze the influence of these parameters on our results in a systematic way. In a first step, we distinguish between a high and a low equity risk premium state. According to Dimson et al. (2006), the long-run risk premium for developed stock markets is roughly 7% per year, this value represents our high risk premium state. They further argue that the equity risk premium might be lower in the years ahead and estimate the annual expected stock market excess return to be 4.5%. This conservative estimate represents our low risk premium state. Following Arnott and Bernstein (2002), we set the risk-free rate (money market rate) to a fixed value of 4.5% per year. Adding this value for the risk-free rate to the different equity risk premiums, we specify an expected stock market return of 9% in the low equity risk premium state and of 11.5% in the high equity risk premium state.

In the second step, we distinguish between a high and a low stock market volatility scenario. For the low volatility state, we assume a value of 20%, which roughly represents the average annual stock market volatility (Dimson et al., 2006). In contrast, we use a stock market volatility of 30% per year for the high volatility state. Combining the two different values for the expected return (μ) and the volatility (σ), we have four different scenarios in our Monte Car-

lo simulations: scenario 1 ($\mu = 9\%$; $\sigma = 20\%$), scenario 2: ($\mu = 11.5\%$; $\sigma = 20\%$), scenario 3: ($\mu = 9\%$; $\sigma = 30\%$), and scenario 4: ($\mu = 11.5\%$; $\sigma = 30\%$). Based on these four scenarios, we simulate continuously compounded stock market returns on the basis of a geometric Brownian motion (Hull, 2009). This return-generating process has the main advantage of fulfilling the assumption of normally distributed returns underlying the synthetic put strategy and the dynamic VaR-strategy.

In order to derive the mean and volatility parameters for the geometric Brownian motion return-generating process, the expected return and volatility estimates are transformed into their corresponding continuously compounded counterparts. We simulate 250 daily stock market returns and implement the portfolio insurance strategies in the same way as discussed in section III.A (i.e., without short sales and without leverage, 10 basis points round-trip transaction costs, a trading filter of 2%, and a protection level of 100%). Moreover, we use the scenario-based estimators for the equity risk premium and the volatility in implementing the dynamic VaR-strategy. Given the stochastic character of the geometric Brownian motion, any generated return path may deviate from these ‘true’ parameters, also implying that we implicitly incorporate estimation risk. The necessary volatility for the synthetic put strategy is estimated in the same way. In order to derive the full distribution of the return, risk, and performance measures, we generate 10,000 simulation runs. The test procedure for deriving statistical significance is identical to the approach in our block-bootstrap simulations (see section III.C).

B. Monte Carlo simulation results

In order to validate the bootstrap results presented in Table 8, we again use a multiplier of $m = 9$ for the CPPI strategy and the TIPP strategy in our Monte Carlo simulations, and the confidence level in the dynamic VaR-strategy is set to $\alpha = 0.95$. All Monte Carlo simulation results are presented in Table 10. In all four simulated market scenarios in Panel A of Table 10, the CPPI strategy dominates the stop-loss trading rule in terms of Omega ratios (comparison 1). Panel B of Table 10 reveals that the differences in Omega ratios are statistically significant for all threshold return levels up to 3% (scenario 3 and 4) and up to 4% (scenario 1 and 2). As in our bootstrap simulations, the main driver for this result is the superior downside protection quality of the CPPI strategy. Comparing the CPPI and the TIPP strategy (comparison 2), the Monte Carlo simulation results are also similar to those of our bootstrap

simulations. While the TIPP strategy exhibits the lowest LPM values of all simulated protection strategies, its return potential is comparatively low. Moreover, the TIPP strategy dominates the CPPI strategy in terms of Omega ratios, but only at the cost of its lower returns. With an increasing threshold level τ , the difference in the Omega ratios decreases. For example, in all four scenarios the difference between the two Omega ratios is no longer statistically significant based on a threshold level of 4% (see Panel B of Table 10).

[Insert Table 10 here]

The Monte Carlo simulation results for the comparison between the CPPI strategy and the synthetic put strategy deviate from those of the bootstrap simulations (comparison 3). Based on the Omega ratio, there is a dominance of the CPPI strategy. While the synthetic put strategy shows higher Omega ratios than the CPPI strategy in the bootstrap simulations for threshold returns of 3% and 4% (see Table 8), this effect no longer shows up in the Monte Carlo simulations. For the synthetic put strategy, normally distributed returns lead to lower LPM1 values, but neither to superior LPM0 values nor to higher returns. In contrast, compared to the bootstrap simulations, the LPM0 and LPM1 values of the CPPI strategy improve noticeably in a Monte Carlo framework. Therefore, the CPPI strategy seems to benefit more from all the assumptions underlying the geometric Brownian motion as the return generating process.

The results for the comparison of the synthetic put strategy and the dynamic VaR-strategy are similar for both the bootstrap and the Monte Carlo simulation (comparison 4). While the dynamic VaR-strategy dominates in terms of Omega ratios for threshold levels of 1% and 2% in our Monte Carlo simulations (in all four scenarios), the dynamic VaR-strategy also generates a significantly higher Omega ratio for a 3% threshold return in our Monte Carlo simulations in scenarios 1 and 2 as in our bootstrap simulations (see Table 8). Comparing the LPM0 and LPM1 values of the dynamic VaR-strategy in Table 8 with the corresponding values in Table 10, we observe superior risk properties for the dynamic VaR-strategy in our Monte Carlo simulations. In all scenarios, the shortfall probabilities (as measured by the LPM0) are at least as good as or even lower than the expected values determined by the confidence level α . From this point of view, both the assumption of normally distributed returns and the negligence of deviating time series feature benefit the dynamic VaR-strategy more than the synthetic put strategy. The risk reduction effect of the dynamic VaR-strategy based on normally

distributed returns may also provide an explanation why this strategy provides higher Omega ratios than the CPPI strategy for all threshold returns ranging from 1% to 3% for all four scenarios (albeit statistical significance is only observable for threshold returns up to 2%). This result differs from our bootstrap simulations, where no statistical significance of the dynamic VaR-strategy is observable against the CPPI strategy in terms of their Omega ratios for all threshold levels. Taken as a whole, however, our Monte Carlo simulation results confirm the bootstrap results for the comparison of the dynamic VaR-strategy with the synthetic put strategy as well as the CPPI strategy. Most importantly, our systematic comparison indicates that normally distributed returns are an important requirement for the protection quality of the dynamic VaR-strategy.

VI. Conclusions

This study presents a systematic comparison of portfolio insurance strategies by using simulation analyses in order to conduct statistical significance tests. In addition to popular strategies, we also apply more recently developed protection strategies, such as the dynamic VaR-strategy. Our comparison of the different portfolio insurance strategies is based on the Omega ratio, which incorporates both the return and risk properties of a given strategy and does not depend on any distributional assumptions. In order to test the statistical significance of differences in Omega ratios, we use a bootstrap-based hypothesis test.

Our systematic comparison of different portfolio insurance strategies considers the following distinguishing characteristics: static versus dynamic; initial wealth versus cumulated wealth protection; model-based versus model-free; and strong floor compliance versus probabilistic floor compliance. Overall, our findings reveal that the classical portfolio insurance strategies synthetic put and CPPI provide superior downside protection compared to the simple stop-loss trading rule, which ultimately results in significantly higher Omega ratios. Analyzing alternative portfolio insurance strategies, neither the TIPP strategy (as an ‘improved’ CPPI strategy) nor the dynamic VaR-strategy (with probabilistic floor compliance) are able to provide substantial improvements over the classical strategies. The attractiveness of the dynamic VaR-strategy compared to the CPPI strategy and the synthetic put strategy strongly depends on the quality of the estimates for the required input parameters, in particular, the equity risk premium. However, if an investor possesses superior forecasting skills, other active (market

timing) strategies may exist which generate higher (risk-adjusted) returns compared to a protected passive stock market investment.

Once an investor has decided to use a protection strategy within his asset allocation strategy, our simulation results indicate an inverse relationship between the protection quality and the return potential of the different portfolio insurance strategies. As expected, strategies with a better downside protection (such as the CPPI strategy and the TIPP strategy with a low multiplier) tend to provide lower returns. Implementing the portfolio insurance strategy that best meets an investor's return and risk preferences, the superiority of a strategy when assessed by the Omega ratio depends on the level of the threshold return required by the investor. With higher threshold returns, strategies with a higher return potential become increasingly attractive. In most instances, the CPPI strategy exhibits a superior ratio of realized gains to losses compared to all other portfolio insurance strategies.

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Table 1: Main properties of portfolio insurance strategies

		Stop-loss	Synthetic Put	CPPI	TIPP	Dynamic VaR
Insurance strategy		Static	Dynamic	Dynamic	Dynamic	Dynamic
Protection target		Initial wealth	Initial wealth	Initial wealth	Cumulated wealth	Initial wealth
Type of insurance		Strong floor compliance	Strong floor Compliance	Strong floor compliance	Strong floor compliance	Probabilistic floor compliance
Model parameters	Fixed	Floor	- Stock price (S) - Strike price (K) - Risk-free rate (r_f) - Time to maturity (T)	- Floor - Multiplier (m)	- Floor - Multiplier (m)	- VaR (Floor) - Risk-free rate (r_f) - Confidence level (α)
	Estimators		Volatility (σ_S)			- Equity risk premium ($\mu_S - r_f$) - Volatility (σ_S)
Model assumptions		No assumptions	Normal distribution	No assumptions	No assumptions	Normal distribution

Table 2: Data description

<i>Panel A: Mean return and volatility</i>		
	Stock market	Money market
Mean return p.a.	7.38%	4.70%
Volatility p.a.	21.53%	0.17%
<i>Panel B: Autocorrelation</i>		
	Value	Q-test statistic
AC(1)	-0.001	0.01
AC(2)	-0.030	7.49 **
AC(3)	-0.021	11.18 **
AC(4)	0.025	16.21 ***
AC(5)	-0.011	17.15 ***
<i>Panel C: Heteroscedasticity</i>		
	Value	Lagrange multiplier test statistic
ARCH(1)	–	315.39 ***
ARCH(3)	–	893.87 ***
<i>Panel D: Skewness, kurtosis, and test for normal distribution</i>		
	Value	Jarque-Bera test statistic
Skewness	-0.32	–
Kurtosis	10.03	–
Normality test	–	17311.29 ***

This table shows descriptive statistics of the German stock and money markets. The data consists of continuously compounded daily returns for the German stock market index DAX and money market rates from the German Bundesbank (average values of the 1-month Frankfurt interbank rate; middle rate) over the sample period from January 1980 to December 2011. The daily average returns in Panel A are annualized by multiplying with 250 (trading days). The volatility of daily returns is transformed into an annual volatility by multiplying with $\sqrt{250}$. The additional analyses in Panels B-D are limited to the stock market returns. AC(.) in Panel B denotes the autocorrelation in returns at the lag shown in brackets. The statistical significance of autocorrelation up to lag five is tested using the Ljung-Box test (Q-test statistic). ARCH(.) in Panel C refers to heteroskedasticity effects at the lag indicated in brackets. The test statistic is based on Engle's (1982) Lagrange multiplier test; the null hypothesis is that there are no heteroskedasticity effects in daily DAX returns at lag one and up to lag three. Panel D reports higher moments of stock returns (skewness and kurtosis). The Jarque-Bera test statistic tests the null hypothesis that stock returns are normally distributed. ***/**/* denotes statistical significance at the 1%, 5%, and 10% level, respectively.

Table 3: Historical simulation results

<i>Panel A: Return, risk, and performance of portfolio insurance strategies</i>						
	Stop-loss	Synthetic put	CPPI ($m = 5$)	TIPP ($m = 5$)	Dynamic VaR ($\alpha = 0.90$)	Stock market
Mean return p.a.	8.97%	8.19%	5.82%	4.88%	7.31%	11.36%
LPM0 ($\tau = 0\%$)	68.10%	28.94%	6.98%	3.48%	28.54%	31.84%
LPM1 ($\tau = 0\%$)	1.06%	0.55%	0.04%	0.02%	0.40%	5.41%
VaR (95%)	-3.73%	-3.02%	-0.12%	0.24%	-2.36%	-30.28%
Skewness	1.58	1.78	2.07	0.68	2.32	-0.08
Min. return	-9.80%	-8.66%	-2.21%	-2.21%	-9.15%	-57.32%
$\Omega(\tau = 1\%)$	5.57	9.14	25.03	39.85	9.32	2.81
$\Omega(\tau = 2\%)$	3.87	5.73	8.89	11.27	5.42	2.54
$\Omega(\tau = 3\%)$	2.92	3.91	4.24	4.33	3.51	2.31
$\Omega(\tau = 4\%)$	2.31	2.83	2.35	1.91	2.46	2.09

Continued...

Table 3: Continued

<i>Panel B: Statistical significance</i>					
	Comparison 1	Comparison 2	Comparison 3	Comparison 4	
	CPPI vs. stop-loss	CPPI vs. TIPP	CPPI vs. synthetic put	Dynamic VaR vs. synthetic put	Dynamic VaR vs. CPPI
Mean return p.a.	-3.15 ***	0.94 ***	-2.37 ***	-0.88 ***	1.49 ***
LPM0($\tau = 0\%$)	-61.22 ***	3.50 ***	-21.96 ***	-0.40	21.56 ***
LPM1($\tau = 0\%$)	-1.02 ***	0.02 ***	-0.51 ***	-0.15 **	0.36 ***
VaR (95%)	3.61 ***	-0.36 ***	2.90 ***	0.66	-2.24 ***
Skewness	0.49 ***	1.39 ***	0.29 **	0.54 ***	0.25 *
Min. return	7.59 ***	0.00 ***	6.45 ***	-0.49	-6.94 ***
$\Omega(\tau = 1\%)$	19.46 ***	-14.82 ***	15.89 ***	0.18	-15.71 ***
$\Omega(\tau = 2\%)$	5.02 ***	-2.38 ***	3.16 ***	-0.31	-3.47 ***
$\Omega(\tau = 3\%)$	1.32 ***	-0.09	0.33	-0.40	-0.73 **
$\Omega(\tau = 4\%)$	0.04	0.44 ***	-0.48 ***	-0.37 *	0.11

This table shows the results of historical simulations using the base-case specifications. The data consists of daily returns for the German stock market index DAX and money market rates from the German Bundesbank (average values of the 1-month Frankfurt interbank rate; middle rate) over the sample period from January 1980 to December 2011. Volatility and risk premium estimates (if necessary) are taken from the year prior to the implementation of the strategy. In order to capture the time series characteristics of the return series, we implement a common block-bootstrap approach drawing 250 subsequently following pairs of daily stock and money market returns with replacement. In particular, selecting $N = 5,000$ coherent data blocks with replacement and applying the different portfolio insurance strategies in each draw provides us with 5,000 yearly returns for each strategy. Panel A presents the return, risk, and performance characteristics of the different portfolio insurance strategies. $m = 5$ denotes the multiplier of the CPPI and the TIPP strategy, and $\alpha = 0.90$ is the confidence level of the dynamic VaR-strategy. In addition to the mean annual returns, the strategies' protection quality is captured by using downside risk measures. Lower partial moments (LPMs) only take negative deviations from the target return (τ) into account. The target return is set equal to zero ($\tau = 0$), and we penalize negative deviations from this target with an exponent of 0 (LPM0) and 1 (LPM1). LPM0 and LPM1 represent the shortfall probability and the expected shortfall, respectively. In addi-

tion, the table reports the empirical VaR (using a confidence level of $\alpha = 95\%$), the skewness, and the minimum annual return as alternative downside risk measures. The Omega measure, defined as the ratio of gains to losses relative to a target return, is used as risk-adjusted performance measure for our systematic comparison of the different portfolio insurance strategies along the distinguishing characteristics as discussed in section II.F (comparisons 1-4 in Panel B). The threshold returns for the Omega ratio are set to 1%, 2%, 3%, and 4% per year, respectively. Panel B reports the differences between the performance and risk measures of each pair of strategies. In order to derive the confidence intervals, we implement a bootstrap as proposed in Efron (1979) using the $N = 5,000$ annual returns from the common block-bootstrap in the first step. Within this second step bootstrap, we again create $N_B = 5,000$ bootstrap resamples, each consisting of 500 annual returns (see Figure I). The reported significance levels are based on the confidence interval with the highest level of significance. If the confidence interval based on a 10% significance level still contains zero, there are no significant differences in the performance/risk measures of the respective pair of strategies. In contrast, if both reported interval boundaries are positive (negative), one strategy features a significantly higher (lower) performance/risk measure than the other at the reported level of statistical significance. ***/**/* denotes statistical significance at the 1%, 5%, and 10% level, respectively.

Table 4: Historical simulation results for the stop-loss strategy with different floor values

	Floor = 99%	Floor = 97.5%	Floor = 95%	Floor = 90%
Mean return p.a.	8.81%	9.51%	10.29%	10.82%
LPM0 ($\tau = 0\%$)	65.40%	59.18%	50.30%	39.68%
LPM1 ($\tau = 0\%$)	1.69%	2.42%	3.28%	4.24%
VaR (95%)	-5.02%	-6.36%	-8.53%	-13.05%
Skewness	1.41	1.15	0.87	0.54
Min. return	-10.57%	-12.02%	-14.40%	-18.26%
$\Omega(\tau = 1\%)$	4.33	3.83	3.46	3.12
$\Omega(\tau = 2\%)$	3.27	3.08	2.93	2.75
$\Omega(\tau = 3\%)$	2.59	2.55	2.52	2.43
$\Omega(\tau = 4\%)$	2.11	2.15	2.18	2.16

With respect to the measures of interest, this table shows the return, risk, and performance values of historical simulations for the stop-loss strategy with different choices for the floor value. The data consists of daily returns for the German stock market index DAX and money market rates from the German Bundesbank (average values of the 1-month Frankfurt interbank rate; middle rate) over the sample period from January 1981 to December 2011. In order to capture the time series characteristics of the return series, we implement a common block-bootstrap approach by drawing 250 subsequently following pairs of daily stock and money market returns with replacement. In particular, selecting $N = 5,000$ coherent data blocks with replacement and applying the different portfolio insurance strategies in each draw provides us with 5,000 yearly returns for each strategy variant. In addition to the mean annual returns, the strategies' protection quality is captured by using downside risk measures. Lower partial moments (LPMs) only take negative deviations from the target return (τ) into account. The target return is set equal to zero ($\tau = 0$), and we penalize negative deviations from this target with an exponent of 0 (LPM0) and 1 (LPM1). LPM0 and LPM1 represent the shortfall probability and the expected shortfall, respectively. In addition, the table reports the values of the empirical VaR (using a confidence level of $\alpha = 95\%$), the skewness, and the minimum annual return as alternative downside risk measures. The Omega measure is defined as the ratio of gains to losses relative to a target return, where the threshold returns are set to 1%, 2%, 3%, and 4% per year, respectively.

Table 5: Historical simulation results for the CPPI and TIPP strategies with different multipliers

	CPPI ($m = 3$)	TIPP ($m = 3$)	CPPI ($m = 7$)	TIPP ($m = 7$)	CPPI ($m = 9$)	TIPP ($m = 9$)
Mean return p.a.	5.30%	4.84%	6.61%	4.97%	7.10%	5.05%
LPM0 ($\tau = 0\%$)	1.98%	1.40%	15.02%	6.54%	25.06%	10.74%
LPM1 ($\tau = 0\%$)	0.01%	0.01%	0.07%	0.04%	0.17%	0.12%
VaR (95%)	0.42%	0.57%	-0.44%	-0.17%	-1.34%	-0.68%
Skewness	0.96	0.34	2.08	0.92	2.18	1.03
Min. return	-1.35%	-1.34%	-2.76%	-2.76%	-3.85%	-3.85%
$\Omega(\tau = 1\%)$	50.36	70.55	17.57	25.36	11.95	15.09
$\Omega(\tau = 2\%)$	12.43	15.21	7.19	8.01	5.89	6.12
$\Omega(\tau = 3\%)$	4.96	5.30	3.94	3.52	3.59	3.09
$\Omega(\tau = 4\%)$	2.35	2.06	2.48	1.79	2.44	1.73

With respect to the measures of interest, this table shows the return, risk, and performance values of historical simulations for the CPPI strategy and TIPP strategy with different choices for the multiplier m (see equation 9). The data consists of daily returns for the German stock market index DAX and money market rates from the German Bundesbank (average values of the 1-month Frankfurt interbank rate; middle rate) over the sample period from January 1981 to December 2011. In order to capture the time series characteristics of the return series, we implement a common block-bootstrap approach by drawing 250 subsequently following pairs of daily stock and money market returns with replacement. In particular, selecting $N = 5,000$ coherent data blocks with replacement and applying the different portfolio insurance strategies in each draw provides us with 5,000 yearly returns for each strategy variant. In addition to the mean annual returns, the strategies' protection quality is captured by using downside risk measures. Lower partial moments (LPMs) only take negative deviations from the target return (τ) into account. The target return is set equal to zero ($\tau = 0$), and we penalize negative deviations from this target with an exponent of 0 (LPM0) and 1 (LPM1). LPM0 and LPM1 represent the shortfall probability and the expected shortfall, respectively. In addition, the table reports the values of the empirical VaR (using a confidence level of $\alpha = 95\%$), the skewness, and the minimum annual return as alternative downside risk measures. The Omega measure is defined as the ratio of gains to losses relative to a target return, where the threshold returns are set to 1%, 2%, 3%, and 4% per year, respectively.

Table 6: Historical simulation results for the dynamic VaR-strategy with different confidence levels

	Confidence level $\alpha \dots$			
	$\alpha = 0.85$	$\alpha = 0.90$	$\alpha = 0.95$	$\alpha = 0.99$
Mean return p.a.	7.21%	6.97%	6.80%	6.13%
LPM0 ($\tau = 0\%$)	32.22%	28.60%	21.44%	10.44%
LPM1 ($\tau = 0\%$)	0.38%	0.39%	0.28%	0.15%
VaR (95%)	-2.31%	-2.37%	-1.80%	-0.55%
Skewness	2.37	2.33	2.33	2.24
Min. return	-7.95%	-9.15%	-8.70%	-9.82%
$\Omega(\tau = 1\%)$	9.10	8.96	11.00	15.39
$\Omega(\tau = 2\%)$	5.27	5.16	5.90	6.93
$\Omega(\tau = 3\%)$	3.43	3.32	3.61	3.78
$\Omega(\tau = 4\%)$	2.40	2.30	2.40	2.31

With respect to the measures of interest, this table shows the return, risk, and performance values of historical simulations for the dynamic VaR-strategy with different choices for the confidence level α (equation 13). The data consists of daily returns for the German stock market index DAX and money market rates from the German Bundesbank (average values of the 1-month Frankfurt interbank rate; middle rate) over the sample period from January 1980 to December 2011. Volatility and risk premium estimates are taken from the year prior to the implementation of the strategy. In order to capture the time series characteristics of the return series, we implement a common block-bootstrap approach by drawing 250 subsequently following pairs of daily stock and money market returns with replacement. In particular, selecting $N = 5,000$ coherent data blocks with replacement and applying the different portfolio insurance strategies in each draw provides us with 5,000 yearly returns for each strategy. In addition to the mean annual returns, the strategies' protection quality is captured by using downside risk measures. Lower partial moments (LPMs) only take negative deviations from the target return (τ) into account. The target return is set equal to zero ($\tau = 0$), and we penalize negative deviations from this target with an exponent of 0 (LPM0) and 1 (LPM1). LPM0 and LPM1 represent the shortfall probability and the expected shortfall, respectively. In addition, the table reports the values of the empirical VaR (using a confidence level of $\alpha = 95\%$), the skewness, and the minimum annual return as alternative downside risk measures. The Omega measure is defined as the ratio of gains to losses relative to a target return, where the threshold returns are set to 1%, 2%, 3%, and 4% per year, respectively.

Table 7: Historical simulation results for the dynamic VaR-strategy with alternative volatility and risk premium estimators

	<i>Panel A: Alternative volatility estimators</i>			
	Exponentially-weighted volatility	GARCH(1,1)	Squared returns	Historical volatility (prior 250 days)
Mean return p.a.	7.06%	7.08%	6.98%	7.00%
LPM0 ($\tau = 0\%$)	23.36%	22.26%	20.22%	19.80%
LPM1 ($\tau = 0\%$)	0.26%	0.26%	0.24%	0.23%
VaR (95%)	-1.86%	-1.72%	-1.41%	-1.41%
Skewness	2.51	2.39	2.38	2.37
Min. return	-9.14%	-8.25%	-8.70%	-8.70%
$\Omega(\tau = 1\%)$	11.39	11.67	12.08	12.27
$\Omega(\tau = 2\%)$	6.03	6.25	6.34	6.41
$\Omega(\tau = 3\%)$	3.71	3.82	3.84	3.88
$\Omega(\tau = 4\%)$	2.50	2.55	2.54	2.56

Continued...

Table 7: Continued

	<i>Panel B: Alternative risk premium estimators</i>			
	Average return over full sample period	Long-run stock market return	Zero risk premium	Historical return (prior 250 days)
Mean return p.a.	6.56%	6.70%	6.05%	6.95%
LPM0 ($\tau = 0\%$)	14.62%	14.96%	12.14%	21.74%
LPM1 ($\tau = 0\%$)	0.07%	0.07%	0.06%	0.28%
VaR (95%)	-0.23%	-0.25%	-0.14%	-1.93%
Skewness	2.14	2.22	1.96	2.26
Min. return	-6.23%	-6.09%	-6.46%	-9.03%
$\Omega(\tau = 1\%)$	19.72	19.73	20.47	11.07
$\Omega(\tau = 2\%)$	8.23	8.29	8.09	5.98
$\Omega(\tau = 3\%)$	4.36	4.42	4.10	3.69
$\Omega(\tau = 4\%)$	2.63	2.69	2.40	2.47

With respect to the measures of interest, this table shows the return, risk, and performance values of historical simulations for the dynamic VaR-strategy with different estimates for the volatility measures and the equity risk premium. The data consists of daily returns for the German stock market index DAX and money market rates from the German Bundesbank (average values of the 1-month Frankfurt interbank rate; middle rate) over the sample period from January 1980 to December 2011. In our baseline specification, volatility and risk premium estimates are taken as second and first moment from the year prior to the implementation of the strategy; these results are reported in the last column of Panels A and B (estimates based on prior 250 days). As alternative volatility estimators, in Panel A we use the exponentially-weighted moving average model, the GARCH(1,1) model, and the average of the squared returns (setting the mean value to zero). For alternative estimates of the equity risk premium, in Panel B we estimate the expected stock market return using the average return over the full sample period and the (fixed) long-run stock market return (9% per year) estimated by Dimson et al. (2006). In another robustness test, we set the equity risk premium equal to zero. In order to capture the time series characteristics of the return series, we implement a common block-bootstrap approach by drawing 250 subsequently following pairs of daily stock and money market returns with replacement. In particular, selecting $N = 5,000$ coherent data blocks with replacement and applying the different portfolio insurance strategies in each draw provides us with 5,000 yearly returns for each strategy variant. In addition to the mean

annual returns, the strategies' protection quality is captured by using downside risk measures. Lower partial moments (LPMs) only take negative deviations from the target return (τ) into account. The target return is set equal to zero ($\tau = 0$), and we penalize negative deviations from this target with an exponent of 0 (LPM0) and 1 (LPM1). LPM0 and LPM1 represent the shortfall probability and the expected shortfall, respectively. In addition, the table reports the values of the empirical VaR (using a confidence level of $\alpha = 95\%$), the skewness, and the minimum annual return as alternative downside risk measures. The Omega measure is defined as the ratio of gains to losses relative to a target return, where the threshold returns are set to 1%, 2%, 3%, and 4% per year, respectively.

Table 8: Robustness checks for historical simulation results

<i>Panel A: Return, risk, and performance of portfolio insurance strategies</i>						
	Stop-loss	Synthetic put	CPPI ($m = 9$)	TIPP ($m = 9$)	Dynamic VaR ($\alpha = 0.95$)	Stock market
Mean return p.a.	8.67%	8.08%	6.95%	5.07%	6.90%	11.50%
LPM0 ($\tau = 0\%$)	68.54%	26.60%	22.86%	9.92%	13.16%	30.16%
LPM1 ($\tau = 0\%$)	1.03%	0.50%	0.15%	0.10%	0.06%	5.05%
VaR (95%)	-3.86%	-2.95%	-0.99%	-0.59%	-0.21%	-28.54%
Skewness	1.58	1.76	2.23	1.04	2.26	-0.11
Min. return	-10.62%	-8.99%	-3.87%	-3.87%	-6.98%	-58.81%
$\Omega(\tau = 1\%)$	5.47	9.68	12.49	16.75	21.81	2.96
$\Omega(\tau = 2\%)$	3.78	5.96	5.99	6.53	8.94	2.68
$\Omega(\tau = 3\%)$	2.83	4.00	3.59	3.21	4.73	2.42
$\Omega(\tau = 4\%)$	2.23	2.86	2.41	1.77	2.88	2.18

Continued...

Table 8: Continued

<i>Panel B: Statistical significance</i>					
	Comparison 1	Comparison 2	Comparison 3	Comparison 4	
	CPPI vs. stop-loss	CPPI vs. TIPP	CPPI vs. synthetic put	Dynamic VaR vs. synthetic put	Dynamic VaR vs. CPPI
Mean return p.a.	-1.72 ***	1.88 ***	-1.13 ***	-1.18 ***	-0.05
LPM0($\tau = 0\%$)	-45.68 ***	12.94 ***	-3.74 **	-13.44 ***	-9.70 ***
LPM1($\tau = 0\%$)	-0.88 ***	0.05 ***	-0.35 ***	-0.44 ***	-0.09 ***
VaR (95%)	2.87 ***	-0.40 ***	1.96 ***	2.74 ***	0.78 ***
Skewness	0.65 ***	1.19 ***	0.47 ***	0.50 ***	0.03
Min. return	6.75 ***	0.00 ***	5.12 ***	2.01 **	-3.11
$\Omega(\tau = 1\%)$	7.02 ***	-4.26 ***	2.81 ***	12.13 ***	9.32 ***
$\Omega(\tau = 2\%)$	2.21 ***	-0.54	0.03	2.98 ***	2.95 ***
$\Omega(\tau = 3\%)$	0.76 ***	0.38	-0.41 **	0.73 ***	1.14 ***
$\Omega(\tau = 4\%)$	0.18	0.64 ***	-0.45 ***	0.02	0.47 ***

This table shows the results of historical simulations using alternative strategy specifications. In particular, in contrast to the base-case specification in Table 3, a multiplier of $m = 9$ (instead of $m = 5$) is used for both the CPPI and the TIPP strategy and a confidence level of $\alpha = 0.95$ (instead of $\alpha = 0.90$) for the dynamic VaR-strategy. Moreover, the dynamic VaR-strategy is implemented using the (fixed) long-run stock market return (9% per year) estimated by Dimson et al. (2006). Volatility estimates (if necessary) are taken from the year prior to the implementation of the strategy. The data consists of daily returns for the German stock market index DAX and money market rates from the German Bundesbank (average values of the 1-month Frankfurt interbank rate; middle rate) over the sample period from January 1980 to December 2011. In order to capture the time series characteristics of the return series, we implement a common block-bootstrap approach by drawing 250 subsequently following pairs of daily stock and money market returns with replacement. In particular, selecting $N = 5,000$ coherent data blocks with replacement and applying the different portfolio insurance strategies in each draw provides us with 5,000 yearly returns for each strategy. Panel A presents the return, risk, and performance characteristics of the different portfolio insurance strategies. In addition to the mean annual returns, the strategies' pro-

tection quality is captured by using downside risk measures. Lower partial moments (LPMs) only take negative deviations from the target return (τ) into account. The target return is set equal to zero ($\tau = 0$), and we penalize negative deviations from this target with an exponent of 0 (LPM0) and 1 (LPM1). LPM0 and LPM1 represent the shortfall probability and the expected shortfall, respectively. In addition, the table reports the values of the empirical VaR (using a confidence level of $\alpha = 0.95$), the skewness, and the minimum annual return as alternative downside risk measures. The Omega measure, defined as the ratio of gains to losses relative to a target return, is used as risk-adjusted performance measure for our systematic comparison of the different portfolio insurance strategies along the distinguishing characteristics as discussed in section II.F (comparisons 1-4 in Panel B). The threshold returns for the Omega ratio are set to 1%, 2%, 3%, and 4% per year, respectively. Panel B reports the differences between the performance and risk measures of each pair of strategies. In order to derive the confidence intervals, we implement a bootstrap as proposed in Efron (1979) using the $N = 5,000$ annual returns from the common block-bootstrap in the first step. Within this second step bootstrap, we again create $N_B = 5,000$ bootstrap resamples, each consisting of 500 annual returns (see Figure I). The reported significance levels are based on the confidence interval with the highest level of significance. If the confidence interval based on a 10% significance level still contains zero, there are no significant differences in the performance/risk measures of the respective pair of strategies. In contrast, if both reported interval boundaries are positive (negative), one strategy features a significantly higher (lower) performance/risk measure than the other at the reported level of statistical significance. ***/**/* denotes statistical significance at the 1%, 5%, and 10% level, respectively.

Table 9: Historical simulation results for the dynamic VaR-strategy and the synthetic put strategy without estimation risk

	Dynamic VaR-strategy with confidence level $\alpha \dots$				Synthetic put
	$\alpha = 0.85$	$\alpha = 0.90$	$\alpha = 0.95$	$\alpha = 0.99$	
Mean return p.a.	4.96%	5.79%	6.86%	7.69%	7.89%
LPM0 ($\tau = 0\%$)	27.12%	17.42%	8.60%	2.20%	21.90%
LPM1 ($\tau = 0\%$)	0.27%	0.21%	0.10%	0.01%	0.20%
VaR 95%	-1.70%	-1.49%	-0.53%	0.37%	-1.32%
Skewness	2.40%	2.18	1.97	2.53	1.68
Min. return	-7.56%	-7.17	-6.67%	-4.50%	-4.12%
$\Omega(\tau = 1\%)$	6.99	10.63	22.43	84.19%	14.35
$\Omega(\tau = 2\%)$	3.49	5.07	9.47	22.89%	7.13
$\Omega(\tau = 3\%)$	2.09	2.88	4.82	9.24%	4.35
$\Omega(\tau = 4\%)$	1.39	1.85	2.85	4.70%	2.96

With respect to the measures of interest, this table shows return, risk, and performance values of historical simulations for the dynamic VaR-strategy with different choices for the confidence level α (equation 13) and the synthetic put strategy assuming in both cases that estimation risk is ignored. In contrast to previous simulations, where the volatility and risk premium estimates are taken from the year prior to the implementation of the strategy, the realized volatility and the realized risk premium are used in any given year. The data consists of daily returns for the German stock market index DAX and money market rates from the German Bundesbank (average values of the 1-month Frankfurt interbank rate; middle rate) over the sample period from January 1981 to December 2011. In order to capture the time series characteristics of the return series, we implement a common block-bootstrap approach by drawing 250 subsequently following pairs of daily stock and money market returns with replacement. In particular, selecting $N = 5,000$ coherent data blocks with replacement and applying the different portfolio insurance strategies in each draw provides us with 5,000 yearly returns for each strategy. In addition to the mean annual returns, the strategies' protection quality is captured by using downside risk measures. Lower partial moments (LPMs) only take negative deviations from the target return (τ) into account. The target return is set equal to zero ($\tau = 0$), and we penalize negative deviations from this target with an exponent of 0 (LPM0) and 1 (LPM1). LPM0 and LPM1 represent the shortfall probability and the expected shortfall, respectively. Moreover, the table reports the values of the empirical VaR (us-

ing a confidence level of $\alpha = 95\%$), the skewness, and the minimum annual return as alternative downside risk measures. The Omega measure is defined as the ratio of gains to losses relative to a target return, where the threshold returns are set to 1%, 2%, 3%, and 4% per year, respectively.

Table 10: Monte-Carlo simulations

<i>Panel A: Return, risk, and performance of portfolio insurance strategies</i>						
	Stop-loss	Synthetic put	CPPI ($m = 9$)	TIPP ($m = 9$)	Dyn. VaR ($\alpha = 0.95$)	Stock market
Scenario 1: Return 9%, volatility 20%						
Mean return p.a.	6.29%	6.29%	6.12%	5.03%	5.35%	9.29%
LPM0 ($\tau = 0\%$)	75.19%	31.48%	8.00%	1.36%	5.01%	34.31%
LPM1 ($\tau = 0\%$)	0.57%	0.32	0.05%	0.01%	0.03%	3.75%
$\Omega(\tau = 1\%)$	5.00%	8.35	15.53	57.53%	22.64	3.02
$\Omega(\tau = 2\%)$	3.06%	4.47	5.86	14.02%	6.99	1.63
$\Omega(\tau = 3\%)$	2.16%	2.81	3.16	4.98%	3.26	2.29
$\Omega(\tau = 4\%)$	1.64%	1.94	2.01	2.14%	1.84	2.00
Scenario 2: Return 11.5%, volatility 20%						
Mean return p.a.	7.69%	7.41%	7.14%	5.37%	6.21%	11.38%
LPM0 ($\tau = 0\%$)	71.06%	28.17%	7.91%	1.33%	5.30%	29.99%
LPM1 ($\tau = 0\%$)	0.53%	0.30%	0.06%	0.01%	0.03%	3.12%
$\Omega(\tau = 1\%)$	6.39%	10.74	20.23	74.89	27.24	4.03
$\Omega(\tau = 2\%)$	3.91%	5.81	7.78	18.08	8.81	3.50
$\Omega(\tau = 3\%)$	2.76%	3.67	4.19	6.36	4.25	3.04
$\Omega(\tau = 4\%)$	2.09%	2.54	2.65	2.72	2.45	2.65
Scenario 3: Return 9%, volatility 30%						
Mean return p.a.	5.37%	5.63%	5.53%	4.75%	4.67%	8.75%
LPM0 ($\tau = 0\%$)	84.99%	48.72%	3.10%	0.35%	2.97%	43.27%
LPM1 ($\tau = 0\%$)	0.92%	0.32%	0.01%	0.00%	0.01%	7.36%
$\Omega(\tau = 1\%)$	3.46	6.07	8.82	35.77	17.07	1.99
$\Omega(\tau = 2\%)$	2.28	3.27	3.69	8.94	4.91	1.82
$\Omega(\tau = 3\%)$	1.68	2.13	2.21	3.49	2.32	1.66
$\Omega(\tau = 4\%)$	1.32	1.53	1.53	1.64	1.35	1.52
Scenario 4: Return 11.5%, volatility 30%						
Mean return p.a.	6.69%	6.62%	6.52%	4.98%	5.28%	11.56%
LPM0 ($\tau = 0\%$)	82.26%	44.97%	2.82%	0.39%	2.56%	38.78%
LPM1 ($\tau = 0\%$)	0.86%	0.30%	0.01%	0.00%	0.01%	6.22%
$\Omega(\tau = 1\%)$	4.38	7.55	11.45	40.82	21.05	2.60
$\Omega(\tau = 2\%)$	2.87	4.07	4.71	10.33	6.07	2.36
$\Omega(\tau = 3\%)$	2.11	2.66	2.79	4.02	2.89	2.15
$\Omega(\tau = 4\%)$	1.65	1.91	1.92	1.89	1.70	1.96

Continued...

Table 10: Continued

<i>Panel B: Statistical significance</i>					
	Comparison 1	Comparison 2	Comparison 3	Comparison 4	
	CPPI vs. stop-loss	CPPI vs. TIPP	CPPI vs. synthetic put	Dyn. VaR vs. synthetic put	Dyn. VaR vs. CPPI
Scenario 1: Return 9%, volatility 20%					
Mean return p.a.	-0.17	1.09 ***	-0.17	-0.94 ***	-0.77 ***
LPM0 ($\tau = 0\%$)	-67.19 ***	6.64 ***	-23.48 ***	-26.47 ***	-2.99 **
LPM1 ($\tau = 0\%$)	-0.52 ***	0.04 ***	-0.27 ***	-0.29 ***	-0.02 *
$\Omega(\tau = 1\%)$	10.53 ***	-42.00 ***	7.18 ***	14.29 ***	7.11 ***
$\Omega(\tau = 2\%)$	2.80 ***	-8.16 ***	1.39 ***	2.52 ***	1.13 **
$\Omega(\tau = 3\%)$	1.00 ***	-1.82 ***	0.35 ***	0.45 ***	0.10
$\Omega(\tau = 4\%)$	0.37 **	-0.13	0.07	-0.10	-0.17
Scenario 2: Return 11.5%, volatility 20%					
Mean return p.a.	-0.55	1.77 ***	-0.27 *	-1.20 ***	-0.93 ***
LPM0 ($\tau = 0\%$)	-63.15 ***	6.58 ***	-20.26 ***	-22.87 ***	-2.61 *
LPM1 ($\tau = 0\%$)	-0.47 ***	0.05 ***	-0.24 ***	-0.27 ***	-0.03 **
$\Omega(\tau = 1\%)$	13.84 ***	-54.66 ***	9.49 ***	16.50 ***	7.01 ***
$\Omega(\tau = 2\%)$	3.87 ***	-10.30 ***	1.97 ***	3.00 ***	1.03 **
$\Omega(\tau = 3\%)$	1.43 ***	-2.17 ***	0.52 ***	0.58 ***	0.06
$\Omega(\tau = 4\%)$	0.56 ***	-0.07	0.11	-0.09	-0.20
Scenario 3: Return 9%, volatility 30%					
Mean return p.a.	0.16	0.78	-0.10	-0.96 ***	-0.86 *
LPM0 ($\tau = 0\%$)	-81.89 ***	2.75 ***	-45.62 ***	-45.75 ***	-0.13
LPM1 ($\tau = 0\%$)	-0.91 ***	0.01 ***	-0.33 ***	-0.31 ***	0.00
$\Omega(\tau = 1\%)$	5.36 ***	-26.95 ***	2.75 ***	11.00 ***	8.25 ***
$\Omega(\tau = 2\%)$	1.41 ***	-5.25 ***	0.42 *	1.64 ***	1.22 ***
$\Omega(\tau = 3\%)$	0.53 **	-1.28 ***	0.08	0.19	0.11
$\Omega(\tau = 4\%)$	0.21	-0.11	0.00	-0.18 **	-0.18
Scenario 4: Return 11.5%, volatility 30%					
Mean return p.a.	-0.17	1.54 ***	-0.10	-1.34 ***	-1.24
LPM0 ($\tau = 0\%$)	-79.44 ***	2.43 ***	-42.15 ***	-42.41 ***	-0.26
LPM1 ($\tau = 0\%$)	-0.85 ***	0.01 ***	-0.29 ***	-0.29 ***	0.00
$\Omega(\tau = 1\%)$	7.07 ***	-29.37 ***	3.90 ***	13.50 ***	9.60 ***
$\Omega(\tau = 2\%)$	1.84 ***	-5.62 ***	0.64 **	2.00 ***	1.36 ***
$\Omega(\tau = 3\%)$	0.68 ***	-1.23 ***	0.13	0.23	0.10
$\Omega(\tau = 4\%)$	0.27	0.03	0.01	-0.21 **	-0.22

With respect to the measures of interest, this table shows the return, risk, and performance values of Monte-Carlo simulations as a robustness test. Based on four scenarios (with given return and volatility assumptions), continuously compounded stock market returns are simulated on the basis of a geometric Brownian motion. The expected return and volatility estimates are transformed into their continuously compounded counterparts. $m = 9$ denotes the multiplier of the CPPI and the TIPP strategy, and $\alpha = 0.95$ is the confidence level of the dynamic VaR-strategy. We simulate 250 daily stock market returns and implement the portfolio insurance strategies. The scenario-based estimators for the equity risk premium and the volatility are used in the implementation of the dynamic VaR-strategy. The required volatility for the synthetic put strategy is estimated in the same way. In order to derive the full distribution of the return, risk, and performance measures, we perform 10,000 simulation runs. Panel A presents the return, risk, and performance characteristics of the different portfolio insurance strategies. In addition to the mean annual returns, the strategies' protection quality is captured by using downside risk measures. Lower partial moments (LPMs) only take negative deviations from the target return (τ) into account. The target return is set equal to zero ($\tau = 0$), and we penalize negative deviations from this target with an exponent of 0 (LPM0) and 1 (LPM1). LPM0 and LPM1 represent the shortfall probability and the expected shortfall, respectively. The Omega measure, defined as the ratio of gains to losses relative to a target return, is used for our systematic comparison of the different portfolio insurance strategies along the distinguishing characteristics as discussed in section II.F (comparisons 1-4 in Panel B). The threshold returns for the Omega ratio are set to 1%, 2%, 3%, and 4% per year, respectively. Panel B reports the differences between the performance and risk measures of each pair of strategies. In order to derive the confidence intervals, we implement a bootstrap as proposed in Efron (1979) using the $N = 5,000$ annual returns from the common block-bootstrap in the first step. Within this second step bootstrap, we again create $N_B = 5,000$ bootstrap resamples, each consisting of 500 annual returns (see Figure I). The reported significance levels are based on the confidence interval with the highest level of significance. If the confidence interval based on a 10% significance level still contains zero, there are no significant differences in the performance/risk measures of the respective pair of strategies. In contrast, if both reported interval boundaries are positive (negative), one strategy features a significantly higher (lower) performance/risk measure than the other at the reported level of statistical significance. ***/**/* denotes statistical significance at the 1%, 5%, and 10% level, respectively.

Figure 1: The historical bootstrap setup

